LEARNER GUIDE

Zero as a placeholder

Place Value · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A zero is worth nothing, but it can still do a job: it **holds a column open**, so the other digits stay where they belong. Take the 0 out of 4,050 and the 4 slides from the thousands into the hundreds: the number becomes 450. Some zeros hold nothing. A zero at the front of a whole number (0450) or at the end of a decimal (3.50) has no digits beyond it to slide, so it can go.

What you are learning to notice

- A zero matters when other digits would slide into new columns without it.
- 4,050 and 450 are different numbers. 0450 and 450 are the same.
- 3.50 and 3.5 are the same number: both are 3 ones and 5 tenths.
- When you write a number from words, give every column a digit, using 0 for empty ones.
- Thousands need four columns; millions need seven.

Moving through it, one step at a time

To write a number from words, work out how many columns it needs (four for thousands, seven for millions). Put each digit you hear into its column, then put a 0 in every column that is still empty. To check whether a zero can go, take it away in your head: if any other digit has to slide into a new column, the zero was needed.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A zero that holds a column

What happens if the 0 in the hundreds column of 4,050 is left out?

4,050

Thousands	Hundreds	Tens	Ones
4	0	5	0

Take the hundreds 0 away

Thousands	Hundreds	Tens	Ones
	4	5	0

The 4 slides from the thousands into the hundreds. The number is now 450.

1. 4,050 is	4 thousands, 0 hundreds, 5 tens, 0 ones
2. Without the hundreds 0, the 4 slides one column right.	4 hundreds
3. Read the new number.	450

4,050 and 450 are different numbers. That zero was holding the hundreds column open.

Example 2 — A zero that holds nothing

Is 3.50 the same number as 3.5?

3.50

Ones		Tenths	Hundredths
3	•	5	0

Take the last 0 away

Ones		Tenths	Hundredths
3	•	5	

No other digit moves. The 3 is still 3 ones and the 5 is still 5 tenths.

1. 3.50 is	3 ones, 5 tenths, 0 hundredths
2. Take the last 0 away. Does any digit move?	no
3. So the value is unchanged.	3.5

3.50 = 3.5. The end zero of a decimal has no digits beyond it to hold in place.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (what each column is worth (Lesson 1) and how digits move (Lesson 2)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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